

September 2026 update

See below the latest updates for the current 21st edition, which are all incorporated in the PDF file of the Kourovka Notebook available at <https://kourovkanotebookorg.wordpress.com/>.

All the updates since the first appearance of the 21st edition are also listed below; the problems with the newest comments or solutions added in September 2026 are marked **NEW**.

***3.46.** Does there exist a group having more than one, but finitely many maximal locally soluble normal subgroups?

B. I. Plotkin

*Yes, such groups exist (P. Monticone, *Preprint*, 2026, https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2026/06/3_46.pdf); see also (W. van Doorn, E. Judin, P. Monticone, D. Morrison, *Preprint*, 2026, <https://arxiv.org/abs/2607.17477>).

NEW ***6.30.** Let G be a residually finite Hopfian group, and let $\widehat{G}$ be its profinite completion. Is $\widehat{G}$ necessarily Hopfian (in topological sense)?

O. V. Mel'nikov

*No, not always (M. Brescia, E. Ingrosso, M. Trombetti, *Preprint*, 2026, <https://arxiv.org/abs/2607.19193>).

NEW ***10.34.** Does there exist a non-soluble finite group which coincides with the product of any two of its non-conjugate maximal subgroups?

Editors' comment: the answer is negative for almost simple groups (T. V. Tikhonenko, V. N. Tyutyanov, *Siberian Math. J.*, **51**, no. 1 (2010), 174–177).

V. S. Monakhov

*No, there are no such groups (R. Sater, *Preprint of 6 August 2026*, <https://arxiv.org/abs/2608.02970>; J. Li, Y. Yang, *Preprint of 19 August 2026*, <https://arxiv.org/abs/2608.19478>).

NEW ***11.9.** (I. I. Pyatetskii-Shapiro). Does there exist a finite non-soluble group G such that the set of characters induced by the trivial characters of representatives of all conjugacy classes of subgroups of G is linearly independent?

Ya. G. Berkovich

*No, there are no such groups (M. Brescia, E. Ingrosso, M. Trombetti, *Preprint*, 2026, <https://arxiv.org/abs/2608.16957>).

NEW ***12.15.** Suppose that, in a finite 2-group G , any two elements are conjugate whenever their normal closures coincide. Is it true that the derived subgroup of G is abelian?

E. A. Golikova, A. I. Starostin

*No, not always (K. Muliarchyk, *Preprint*, 2026, https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2026/08/kourovka_12_15_counterexample.pdf).

NEW ***13.23.** Let the group G have a finite normal series with infinite cyclic factors (containing of course G and $\{1\}$). Is it true that G has a non-trivial outer automorphism?

F. de Giovanni

*No, not always (M. Brescia, E. Ingrosso, M. Trombetti, *Preprint*, 2026, <https://arxiv.org/abs/2607.11775>).

NEW *14.23. Let F_n be a free group with basis $\{x_1, \dots, x_n\}$, and let $|\cdot|$ be the length function with respect to this basis. For $\alpha \in \text{Aut } F_n$ we put $\|\alpha\| = \max\{|\alpha(x_1)|, \dots, |\alpha(x_n)|\}$. Is it true that there is a recursive function $f : \mathbb{N} \rightarrow \mathbb{N}$ with the following property: for any $\alpha \in \text{Aut } F_n$ there is a basis $\{y_1, \dots, y_k\}$ of $\text{Fix}(\alpha) = \{x \mid \alpha(x) = x\}$ such that $|y_i| \leq f(\|\alpha\|)$ for all $i = 1, \dots, k$? *O. V. Bogopolski*

*No, this is not true if the function is required to work simultaneously for all ranks (O. Kharlampovich, *Preprint*, 2026, https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2026/08/kourovka_14_23_uniform_counterexample.pdf), but this is true for each given rank (O. Kharlampovich, *Preprint*, 2026, https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2026/08/kourovka_14_23_17_32_note_with_problem_authors.pdf).

*14.54. Let $k(G)$ denote the number of conjugacy classes of a finite group G . Is it true that $k(G) \leq |N|$ for some nilpotent subgroup N of G ? This is true if G is simple; besides, always $k(G) \leq |S|$ for some soluble subgroup $S \leq G$.

M. W. Liebeck, L. Pyber

*No, not always (S. Tertooy, *Preprint*, 2026, <https://arxiv.org/abs/2606.21404>).

NEW *14.85. Suppose that an endomorphism φ of a free metabelian group of rank r takes every primitive element to a primitive one. Is φ necessarily an automorphism? This is true for $r \leq 2$.

E. I. Timoshenko, V. Shpilrain

*Yes, it is (V. Ionin, A. Semidetnov, *Preprint*, 2026, <https://arxiv.org/pdf/2608.29219>).

NEW *15.26. A *partition* of a group is a representation of it as a set-theoretic union of some of its proper subgroups (*components*) that intersect pairwise trivially. Is it true that every nontrivial partition of a finite p -group has an abelian component?

Ya. G. Berkovich

*Yes, it is true (G. Cat, *Preprint*, 2026, https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2026/08/kourovka_15-26-ai.pdf).

15.42. Is it true that the group algebra $k[F]$ of R. Thompson's group F (see 12.20) over a field k satisfies the Ore condition, that is, for any $a, b \in k[F]$ there exist $u, v \in k[F]$ such that $au = bv$ and either u or v is nonzero? If the answer is negative, then F is not amenable.

Comment of 2026: If the answer is positive, then F is amenable (so problem 15.42 is equivalent to 12.20), because the “if and only if” statement holds for groups satisfying Kaplansky's zero-divisor conjecture (D. Kielak, *Preprint*, 2026, <https://arxiv.org/pdf/1605.09133v2>), and F does satisfy it.

V. S. Guba

NEW *16.11. Let G be a finite p -group. Does there always exist a finite p -group H such that $\Phi(H) \cong [G, G]$?

Ya. G. Berkovich

*No, not always (V. Ionin, A. Semidetnov, *Preprint*, 2026, <https://arxiv.org/pdf/2608.29219>).

NEW ***17.32.** Is the following analogue of the Cayley–Hamilton theorem true for the free group F_n of rank n : If $w \in F_n$ and $\varphi \in \text{Aut } F_n$ are such that $\langle w, \varphi(w), \dots, \varphi^n(w) \rangle = F_n$, then $\langle w, \varphi(w), \dots, \varphi^{n-1}(w) \rangle = F_n$? O. V. Bogopolski

*No, it is not true ((V. Ionin, A. Semidetnov, *Preprint*, 2026, <https://arxiv.org/pdf/2608.29219>); O. Kharlampovich, *Preprint*, 2026, https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2026/08/kourovka_14_23_17_32_note_with_problem_authors.pdf).

NEW ***17.47.** Let G be a nilpotent group in which every coset $x[G, G]$ for $x \notin [G, G]$ coincides with the conjugacy class x^G . Is there a bound for the nilpotency class of G ?
If G is finite, then its class is at most 3 (R. Dark, C. Scoppola, *J. Algebra*, **181**, no. 3 (1996), 787–802). S. H. Ghate, A. S. Muktibodh

*No, there is no such bound (V. Ionin, A. Semidetnov, *Preprint*, 2026, <https://arxiv.org/pdf/2608.29219>).

NEW ***17.75.** Can the Monster M act on a nontrivial finite 3-group V so that all elements of M of orders 41, 47, 59, 71 have no nontrivial fixed points in V ? V. D. Mazurov

*No, it cannot; moreover, M cannot act on a 3-group in such a way that an element of order 41 has no nontrivial fixed points, as shown in § 3 of (M. Lee, T. Popiel, *J. Group Theory*, **26** (2023), 193–205).

***17.99.** Consider the group $B = \langle a, b \mid (bab^{-1})a(bab^{-1})^{-1} = a^2 \rangle$ introduced by Baumslag in 1969. The same relation is satisfied by the functions $f(x) = 2x$ and $g(x) = 2^x$ under the operation of composition in the group of germs of monotonically increasing to ∞ continuous functions on $(0, \infty)$, where two functions are identified if they coincide for all sufficiently large arguments. Is the representation $a \rightarrow f, b \rightarrow g$ of the group B faithful? A. Yu. Olshanskii

*No, it is not (C.-F. Nyberg-Brodda, *Preprint*, 2026, <https://arxiv.org/abs/2606.27408>).

NEW ***17.102.** We say that two subsets A, B of an infinite group G are *separated* if there exists an infinite subset X of G such that $1 \in X$, $X = X^{-1}$, and $XAX \cap B = \emptyset$. Is it true that any two disjoint subsets A, B of an infinite group G satisfying $|A| < |G|$, $|B| < |G|$ are separated? This is so if A, B are finite, or G is Abelian. I. V. Protasov

*No, not always (S. Wan, *Preprint*, 2026, <https://arxiv.org/abs/2608.00504>).

NEW ***18.16.** Is it true that any definable endomorphism of any ordered abelian group is of the form $x \mapsto rx$, for some rational number r ? O. V. Belegadek

*Yes, it is true (H. Ji, F. Xu, D. Qiu, *Preprint*, 2026, <https://arxiv.org/abs/2608.17861>).

***18.50.** Let $n \in \mathbb{N}$. Is it true that for every $k \in \{1, \dots, n!\}$ there is some group G and pairwise distinct elements $g_1, \dots, g_n \in G$ such that the set $\{g_{\sigma(1)} \cdots g_{\sigma(n)} \mid \sigma \in S_n\}$ of all products of the g_i obtained by permuting the factors has cardinality k ? S. Kohl

*Yes, it is true (P. Monticone, *Preprint*, 2026, https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2026/05/18_50.pdf; see also (W. van Doorn, E. Judin, P. Monticone, D. Morrison, *Preprint*, 2026, <https://arxiv.org/abs/2607.17477>); see also S. Kohl’s post at <https://mathoverflow.net/questions/123890/numbers-of-distinct-products-obtained-by-permuting-the-factors/511354#511354>).

***19.25.** Let G and H be finite groups of the same order with $\sum_{g \in G} \varphi(|g|) = \sum_{h \in H} \varphi(|h|)$, where φ is the Euler totient function. Suppose that G is simple. Is H necessarily simple?

B. Curtin, G. R. Pourgholi

*No, it need not be (P. Monticone, *Preprint*, 2026, https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2026/05/19_25.pdf); see also (W. van Doorn, E. Judin, P. Monticone, D. Morrison, *Preprint*, 2026, <https://arxiv.org/abs/2607.17477>).

Archive 19.35. Let G be a finite group of order n . Is it true that for every factorization $n = a_1 \cdots a_k$ there exist subsets $A_1, \dots, A_k$ such that $|A_1| = a_1, \dots, |A_k| = a_k$ and $G = A_1 \cdots A_k$?

M. H. Hooshmand

No, it is not true. A counterexample with $k = 3$ is given by the alternating group on 4 letters $G = A_4$ and $(a_1, a_2, a_3) = (2, 3, 2)$. (G. M. Bergman, *Letter of 19 December 2019*, https://math.berkeley.edu/~gbergman/papers/gp_factzn.pdf.) Counterexamples for all $k \geq 3$ are constructed in (R. McCulloch, *Preprint*, 2026, <https://arxiv.org/abs/2607.20569>).

NEW

***19.41.** Let ϕ be an automorphism of a free group F_n of finite rank, and let $F_n \rtimes_{\phi} \mathbb{Z}$ be the split extension of F_n by $\mathbb{Z}$ with $\mathbb{Z}$ acting as $\langle \phi \rangle$. Is the group $F_n \rtimes_{\phi} \mathbb{Z}$ conjugacy separable?

I. Kapovich

*Yes, it is (F. Dahmani, S. Hughes, M. Kudlinska, N. Touikan, *Preprint*, 2026, <https://arxiv.org/abs/2509.22346v2>).

19.57. What are the non-abelian composition factors of a finite group G in which every maximal subgroup is simple or p -nilpotent for some fixed odd prime $p \in \pi(G)$?

Comment of 2026: These factors are determined in the cases where G is non-simple, or alternating, or sporadic simple (A. Beltrán, F. Viudez, *Preprint*, 2026, <https://arxiv.org/abs/2607.06434>).

V. S. Monakhov, V. N. Tyutyanov

NEW

19.58. What are the non-abelian composition factors of a finite group in which every maximal subgroup is simple or p -decomposable for some fixed odd prime $p \in \pi(G)$?

Comment of 2026: These factors are determined in the cases where G is non-simple, or alternating, or sporadic simple (A. Beltrán, F. Viudez, *Preprint*, 2026, <https://arxiv.org/abs/2607.06434>).

V. S. Monakhov, V. N. Tyutyanov

NEW

***19.94.** By definition two elements g_1 and g_2 of a free metabelian group F with basis $\{x_1, \dots, x_n\}$ have the same distribution if the equations $g_1(x_1, \dots, x_n) = g$ and $g_2(x_1, \dots, x_n) = g$ have the same number of solutions in any finite metabelian group G for every $g \in G$. Is it true that elements g_1 and g_2 have the same distribution if and only if they are conjugate by some automorphism of F ?

E. I. Timoshenko

*No, it is not true (V. Ionin, A. Semidetnov, *Preprint*, 2026, <https://arxiv.org/pdf/2608.29219>).

NEW

***20.2.** Are there any nonabelian simple groups of Lie type which are totally 3-closed? There are exactly six nonabelian simple totally 2-closed groups — all are sporadic groups, the largest being the Monster.

M. Arezoomand, M. A. Iranmanesh, C. E. Praeger, G. Tracey

*Yes, there are (T. Gong, M. R. Zeng, Y. Yang, *Preprint*, 2026, <https://arxiv.org/abs/2608.17878>).

NEW

20.71. (J. Lauri). A *card* of a finite simple undirected graph G of order $n = |V(G)|$ is an induced subgraph of order $n - 1$. For a connected graph G with k isomorphism types of cards, can $\text{Aut}(G)$ have more than k orbits on the vertex set $V(G)$

a) for $k = 2$?

b) for $k = 3$?

NEW *c) for $k = 4$?

If a graph G has k isomorphism types of cards, then the group $\text{Aut}(G)$ of automorphisms of G has obviously at least k orbits on $V(G)$. It is known that if all cards are mutually isomorphic, then $\text{Aut}(G)$ is transitive on $V(G)$. Examples are known of graphs G with 5 isomorphism types of cards for which $\text{Aut}(G)$ has 6 orbits on $V(G)$, and of graphs with 6 isomorphism types for which $\text{Aut}(G)$ has 7 orbits on $V(G)$.

V. Pannone

*c) Yes, it can for $k = 4$ (J. Prajapati, *Preprint*, 2026, <https://github.com/infinityscroll/kourovka-20-71>).

NEW ***20.109.** A *skew brace* is a set B equipped with two operations $+$ and $\cdot$ such that $(B, +)$ is an additively written (but not necessarily abelian) group, $(B, \cdot)$ is a multiplicatively written group, and $a \cdot (b + c) = ab - a + ac$ for any $a, b, c \in B$.

Is there a finite skew brace with perfect additive group and non-perfect almost simple multiplicative group?

C. Tsang

*Yes, there is (M. Di Matteo, M. Ferrara, *Preprint*, 2026, <https://arxiv.org/abs/2607.14059>

***20.125.** Does there exist a non-abelian group G and a Rota–Baxter operator $B : G \rightarrow G$ such that B is surjective but not injective?

V. G. Bardakov

*Yes, it does exist (P. Monticone, *Preprint*, 2026, https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2026/03/20_125-1.pdf); see also (W. van Doorn, E. Judin, P. Monticone, D. Morrison, *Preprint*, 2026, <https://arxiv.org/abs/2607.17477>).

21.5. Let p be a prime. Let G be a transitive subgroup of the group of finitary permutations $\text{FSym}(\Omega)$ of a set Ω , let N be a normal subgroup of G , and let S be a transitive Sylow p -subgroup of G .

NEW *a) Is it true that $S \cap N$ is a Sylow p -subgroup of N ?

NEW *b) Is it true that SN/N is a Sylow p -subgroup of G/N ?

c) Are any two transitive Sylow p -subgroups of G locally conjugate in G ?

Two subgroups X, Y of a group G are said to be *locally conjugate* if there is a locally inner automorphism φ of G such that $X^\varphi = Y$. An automorphism φ of G is said to be *locally inner* if for every finite subset $A \subseteq G$ there is an element $g = g(A) \in G$ such that $a^\varphi = g^{-1}ag$ for all $a \in A$.

A. O. Asar

*a) Yes, it is true (P. Monticone, *Preprint*, 2026, https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2026/08/21_5.pdf).

*b) Yes, it is true (P. Monticone, *Preprint*, 2026, https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2026/08/21_5.pdf).

***21.8.** As in 17.57, let $r(m) = \{r + km \mid k \in \mathbb{Z}\}$ for integers $0 \leq r < m$; for $r_1(m_1) \cap r_2(m_2) = \emptyset$ let the *class transposition* $\tau_{r_1(m_1), r_2(m_2)}$ be the involution which interchanges $r_1 + tm_1$ and $r_2 + tm_2$ for each integer t and fixes everything else, and let $\text{CT}(\mathbb{Z})$ be the group generated by all class transpositions.

Let CT_k be the subgroup of $\text{CT}(\mathbb{Z})$ generated by the class transpositions $\tau_{r_1(k), r_2(k)}$ for $0 \leq r_1 \neq r_2 < k$. Since $\tau_{r_1(k), r_2(k)}$ permutes the residue classes modulo k , the group CT_k is isomorphic to the symmetric group S_k . Let $\text{CT}_{(k)} = \langle \text{CT}_2, \text{CT}_3, \dots, \text{CT}_k \rangle$. Is it true that for $k > 3$ the group $\text{CT}_{(k)}$ is isomorphic to the symmetric group S_N , where N is the least common multiple of the numbers $2, 3, \dots, k$?

V. G. Bardakov, A. L. Iskra

*Yes, it is true (Junyao Pan, *Preprint of 14 April 2026*, <https://arxiv.org/abs/2604.12553>; P. Monticone, *Preprint of 30 April 2026*, https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2026/05/21_8-1.pdf); see also (W. van Doorn, E. Judin, P. Monticone, D. Morrison, *Preprint*, 2026, <https://arxiv.org/abs/2607.17477>).

***21.10.** We call a group presentation *finite* if it represents a finite group. We say that a presentation is *just finite* if it is finite and is no longer finite on removal of any relation from it. Is it true that every finite group has a just finite presentation?

Note that if a group has a balanced presentation, then it is just finite. A similar argument can be applied for some p -groups using the Golod–Shafarevich inequality.

Y. Barnea

*Yes, it is true (M. Lackenby, *Preprint*, 2026, <https://arxiv.org/abs/2605.10402>).

***21.12.** Suppose that $\mathcal{U}$ is a nonprincipal ultrafilter on ω , and B is a group such that every element $b \in B$ belongs to a subgroup of B that is a homomorphic image of $\mathbb{Z}^\omega/\mathcal{U}$. Must B then be a homomorphic image of an ultraproduct group $(\prod_{i \in \omega} G_i)/\mathcal{U}$ for some groups G_i ?

This is Question 19 in (G. M. Bergman, *Pacific J. Math.*, **274** (2015) 451–495). An affirmative answer would imply that every torsion group was such a homomorphic image for every $\mathcal{U}$, and so would give positive answers to both parts of 21.11.

G. M. Bergman

*No, it need not (S. M. Corson, *J. Algebra*, **681** (2025), 306–317).

***21.14.** Suppose α is an endomorphism of a group G such that for every group H and every homomorphism $f : G \rightarrow H$, there exists an endomorphism β_f of H such that $\beta_f f = f\alpha$. Must α then be either an inner automorphism of G or the trivial endomorphism? This is Question 5 in (G. M. Bergman, *Publ. Matem.*, **56** (2012), 91–126).

G. M. Bergman

*Yes, it must (F. Fournier-Facio, *Preprint*, 2026, <https://arxiv.org/abs/2604.05728>).

***21.15.** Suppose B is a subgroup of the symmetric group S_Ω on an infinite set Ω . Will the amalgamated free product $S_\Omega *_B S_\Omega$ of two copies of S_Ω with amalgamation of B be embeddable in S_Ω ? This is a weakened form of the group case of Question 4.4 in (G.M. Bergman, *Indag. Math.*, **18** (2007), 349–403).

It is known that $S_\Omega *_B S_\Omega$ need not be so embeddable by a map respecting B (*Algebra Number Theory*, **3** (2009), 847–879, § 10).

G. M. Bergman

*No, not necessarily: take Ω countably infinite, B a subgroup of S_Ω which is not Borel, and $\phi : S_\Omega *_B S_\Omega \rightarrow S_\Omega$ an injective group homomorphism. Let ϕ_1 be the restriction of ϕ to the first copy of S_Ω and ϕ_2 that to the second. Each ϕ_i is continuous (A.S. Kechris, C. Rosendal, *Proc. Lond. Math. Soc.*, **94**, no. 2 (2007), 302–350), so $\text{im}(\phi_i)$ is Borel in S_Ω (as a continuous injective image of a Polish space). Now $B = \phi_1^{-1}(\phi(B)) = \phi_1^{-1}(\text{im}(\phi_1) \cap \text{im}(\phi_2))$ is Borel, a contradiction. (S.M. Corson, *Letter of 26 January 2026*; see also <https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2026/03/solution-of-21.15.pdf>.)

***21.18.** Suppose that G is a finite group, and A_1, A_2, A_3 are subsets of G such that the multiplication map $A_1 \times A_2 \times A_3 \rightarrow G$ is bijective. Must the subgroup $\langle A_2 \rangle$ generated by A_2 have order divisible by the cardinality $|A_2|$? This is Question 8 in (G.M. Bergman, *J. Iranian Math. Soc.*, **1** (2020), 157–161).

It is known (*ibid.*) that the corresponding statement is true for the subgroups $\langle A_1 \rangle$ and $\langle A_3 \rangle$. Moreover, $|A_2|$ will at least divide the order of the least subgroup containing A_2 and closed under conjugation by members of A_1 , and similarly of the least subgroup containing A_2 and closed under conjugation by members of A_3 .

G. M. Bergman

*No, it need not (M.I. Kabenyuk, *Preprint*, 2021, <https://arxiv.org/abs/2102.08605>).

***21.24.** For a finite group G , the *power graph* $\mathcal{P}(G)$ is the graph with vertex set G and edges $\{x, y\}$ for all $x \neq y \in G$ such that either $x \in \langle y \rangle$ or $y \in \langle x \rangle$. Is it true that, for every finite group G , if $\mathcal{P}(G)$ is a cograph, then $\mathcal{P}(G)$ is chordal? Cf. 21.23.

This holds if every element of G has prime power order (D. Bubboloni, F. Fumagalli, C.E. Praeger, *Preprint*, 2025, <https://arxiv.org/abs/2510.18073>) and if G is a nonabelian simple group (J. Cameron, P. Manna, R. Mehatari, *J. Algebra*, **591** (2022), 59–74; J. Brachter, E. Kaja, *J. Algebr. Comb.*, **58** (2023), 1095–1124).

D. Bubboloni, F. Fumagalli, C. E. Praeger

*Yes, it is true (M. Rundström, *Preprint of 30 January 2026*, <https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2026/04/21.24-runds.pdf>; P. Monticone, *Preprint of 29 March 2026*, https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2026/04/21_24-1.pdf); see also (W. van Doorn, E. Judin, P. Monticone, D. Morrison, *Preprint*, 2026, <https://arxiv.org/abs/2607.17477>).

***21.43.** *Conjecture:* Suppose that for a fixed positive integer k at least half of the elements of a finite group G have order k . Then G is solvable.

M. Deaconescu

*This is not always true. In a direct product of N copies of A_5 there are 60^N elements, while the number of elements of order exactly 30 is $60^N - 45^N - 40^N - 36^N + 25^N + 21^N + 16^N - 1$. As $N \rightarrow \infty$, the ratio of elements of order 30 converges to 1. (L. Tae Young, *Letter of 9 January 2026*; see also (R. McCulloch, L. Tae Young, *Preprint*, 2026, <https://arxiv.org/abs/2602.19340>)).

***21.58.** We say that a product $XY = \{xy \mid x \in X, y \in Y\}$ of two subsets X, Y of a group G is *direct* if for every $z \in XY$ there are unique $x \in X, y \in Y$ such that $z = xy$. Is there an infinite group G such that every subset $A \subseteq G$ satisfies the following property: all the maximal subsets B for which the product AB is direct have the same cardinality?

Note that for checking the property for a given infinite group G , it suffices to consider only those subsets $A \subseteq G$ for which $|A| = |G \setminus A|$. Indeed, the property is equivalent to $A^{-1}A \cap BB^{-1} = \{1\}$ and $A^{-1}AB = G$, and these imply $|G| = |A||B|$, since G is infinite. Now, if $|A| < |G \setminus A|$, then $|A| < |G|$, and so $|B| = |G|$; and if $|A| > |G \setminus A|$, then $A^{-1}A = G$, and so $|B| = 1$, for all B satisfying the property.

M. H. Hooshmand

*No, there are no such groups (M. I. Kabenyuk, *Preprint*, February 2026, <https://arxiv.org/abs/2602.22876>; M. H. Hooshmand, *Preprint*, April 2026, <https://arxiv.org/abs/2604.08724>, Remark 1.16).

21.59. For a finite group G , let $\chi_1(G)$ denote the totality of the degrees of all irreducible complex characters of G with allowance for their multiplicities. Suppose that H is a finite group with $\chi_1(H) = \chi_1(G)$.

NEW a) If G is an almost simple group, must H be isomorphic to G ?

*b) If G is a quasisimple group, must H be isomorphic to G ?

This is true if G is a simple group (see 11.8(a) in Archive).

A. Iranmanesh, F. Shirjian

*b) Yes, it must, by Theorem B in (C. Bessenrodt, H. N. Nguyen, J. B. Olsson, H. P. Tong-Viet, *Algebra Number Theory*, **9**, no. 3 (2015), 601–628).

21.67. (*With correct reference to B. Hartley's preprint.*) Suppose that φ is an automorphism of a finite soluble group G , and let r be the (Prüfer) rank of the fixed-point subgroup $C_G(\varphi)$. Is the Fitting height of G bounded terms of $|\varphi|$ and r ?

This is known to be true in the case where $|\varphi|$ is a product of at most two prime powers (B. Hartley, *Preprint*, 1994, <https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2025/08/hartley94-prepr-mims.pdf>). An affirmative answer to this question would imply an affirmative answer to 13.8(b). *E. I. Khukhro*

NEW ***21.87.** Assume that a finite group G has a family of d -generator subgroups whose indices have no common divisor. Is it true that G can be generated by $d+1$ elements?

The answer is positive if G is solvable (L. G. Kovács, H.-S. Sim, *Indag. Math.*, **2** (1991), 229–232). For an arbitrary finite group G it is proved that G can be generated by $d+2$ elements (A. Lucchini, *Commun. Algebra*, **28**, no. 4 (2000), 1875–1880).

A. Lucchini

*Yes, it is true (J. DeCaro, *Preprint*, July 2026, <https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2026/08/21-97-decaro-op-0717-preprint.pdf>); R. Sater, *Preprint*, August 2026, <https://arxiv.org/abs/2608.12432>).

NEW ***21.88.** Is there a finite non-abelian group G of odd order, with $k(G)$ conjugacy classes, such that $k(G)/|G| = 1/17$?

D. MacHale

*No, there are no such groups (B. Beyer de Ryke, *Preprint*, 2026, <https://arxiv.org/abs/2608.03003>).

***21.97.** (With a new version of the preprint with solution.) (M. Tărnăuceanu). Is it true that for every positive rational number r there exists a finite group G such that $|\operatorname{Aut}(G)|/|G| = r$?

A similar question is answered in the positive for graphs, monoids, partial groups, and posets (R. Molinier, *Preprint*, 2025, <https://arxiv.org/abs/2504.21059>). It is also known that the set $\{|\operatorname{Aut}(G)|/|G| \mid G \text{ is a finite abelian group}\}$ is dense in $[0, +\infty)$ (M. Tărnăuceanu, *Elemente Math.* (2025), <https://ems.press/journals/em/articles/14298544>).

R. Molinier

NEW

*Yes, it is true (S. Sureaux, *Preprint*, 2026, <https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2026/08/21.97-solution.pdf>).

***21.117.** (a) Does there exist a finitely generated simple group that is of exponential growth but not of uniformly exponential growth?

(b) Does there exist a finitely generated hereditarily just-infinite group that is of exponential growth but not of uniformly exponential growth?

E. Schesler

*Yes, Thompson's group V provides an affirmative answer to both questions (R. Sauer, E. Schesler, *Preprint*, 2026, <https://arxiv.org/abs/2605.30163>).

***21.134.** For a finite group G , let the *type* of G be the function on positive integers whose value at n is the number of solutions of the equation $x^n = 1$ in G .

a) Is it true that a group having the same type as a group with trivial solvable radical must also have trivial solvable radical? Note that there are solvable and nonsolvable groups with the same type (see 12.37).

b) Is it true that a group having the same type as an almost simple group must be isomorphic to it? This is true for a group having the same type as a simple group, as follows from the affirmative answer to 12.39 (in Archive).

A. V. Vasil'ev

*It turned out that a negative answer to both questions had already been given by J. G. Thompson; see, for example, (Y. Li, W. Shi, *Ric. Mat.*, **74**, no. 1 (2025), 559–563) (*Letter of A. V. Vasil'ev of 18 May 2026*).

NEW

***21.137.** If the p -th powers in a finite p -group form a subgroup, must that subgroup be powerful? That is, for $p \neq 2$, if the p -th powers in a p -group of exponent p^2 form a subgroup, must that subgroup be abelian? For a 2-group of exponent 8, if the squares form a subgroup, must that subgroup be abelian?

L. Wilson

*No, it need not be. For $p = 3$ a counterexample is constructed in (K. Muliarchyk, *Preprint*, 2026, https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2026/08/kourovka_21_137_counterexample.pdf). For $p = 2$, the Sylow 2-subgroup of S_8 has exponent 8, its squares form a subgroup, and that subgroup $C_2 \times D_8$ is nonabelian and not powerful (*Letter of A. Chang of 1 August 2026*).

NEW

***21.142.** A group G is said to be *invariably generated* by a and b if G is generated by the conjugates a^g, b^h for every g, h . Let $p \neq q$ be fixed primes. Does every finite group embed into a finite group invariably generated by an element of order p and an element of order q ?

P. A. Zalesskii

*No, for any two primes $p \neq q$ there is a finite group that does not embed into a finite group invariably generated by an element of order p and an element of order q (T. Gong, M. R. Zeng, Y. Yang, *Preprint of 1 August 2026*, <https://arxiv.org/abs/2608.00703>). Examples for $p = 2, q = 3$ are also constructed in (I. Capdeboscq, C. Parker, *Preprint of 4 August 2026*, <https://arxiv.org/abs/2608.03935>).

***21.147.** (V. M. Kopytov, N. Ya. Medvedev). A subgroup H of a right-orderable group G is said to be *right-relatively convex* if it is convex under some right ordering on G . Is the lattice of right-relatively convex subgroups of a right-orderable group always a sublattice of the lattice of its subgroups?

A. V. Zenkov

*No, not always (P. Monticone, *Preprint*, 2026, https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2026/06/21_147.pdf); see also (W. van Doorn, E. Judin, P. Monticone, D. Morrison, *Preprint*, 2026, <https://arxiv.org/abs/2607.17477>).

21.149. (*With a more precise statement.*) (V. M. Kopytov, N. Ya. Medvedev). Are there order automorphisms of Dlab groups that are not induced by conjugation by elements of a (possibly bigger) Dlab group?

A. V. Zenkov

***21.150.** Let G be an extension of a normal elementary abelian subgroup A by an elementary abelian group $B \cong G/A$ such that A contains an element a with $C_B(a) = 1$. Is it true that the rank of the subgroup $Z(\langle a, B \rangle) \cap (\langle a, B \rangle)'$ is at most the rank of B ?

V. I. Zenkov

*No, not always. Let $A = \mathbb{F}_3[x, y]/I^3$ be the additive group of the quotient of the polynomial algebra $\mathbb{F}_3[x, y]$ by the ideal I^3 , where I is the ideal generated by x, y . Let B be the group of automorphisms of A generated by multiplication by $1 + x$ and $1 + y$. Let $G = A \rtimes B$. For $a = 1 \in A$ we have $C_B(a) = 1_B$. For $H = \langle a, B \rangle$ we have $Z(H) \cap H' = \langle x^2, xy, y^2 \rangle$, which has rank 3, while the rank of B is 2. (P. Monticone, *Letter of 21 March 2026*); see also (P. Monticone, *Preprint*, 2026, https://kourovkanotebookorg.wordpress.com/wp-content/uploads/2026/03/solution_21_150-3.pdf) and (W. van Doorn, E. Judin, P. Monticone, D. Morrison, *Preprint*, 2026, <https://arxiv.org/abs/2607.17477>).